(continued from previous page)

```
1 + -- + ---- + O\x /
    2     24
```

Exercises

1. Calculate $\lim_{x \to 0} \sin(x)/x$
2. Calculate the derivative of $log(x)$ for x.

16.3.4 Integration

SymPy has support for indefinite and definite integration of transcendental elementary and special functions via `integrate()` facility, which uses the powerful extended Risch-Norman algorithm and some heuristics and pattern matching. You can integrate elementary functions:

```
>>> sym.integrate(6 * x ** 5, x)
 6
x
>>> sym.integrate(sym.sin(x), x)
-cos(x)
>>> sym.integrate(sym.log(x), x)
x*log(x) - x
>>> sym.integrate(2 * x + sym.sinh(x), x)
 2
x  + cosh(x)
```

Also special functions are handled easily:

```
>>> sym.integrate(sym.exp(-x ** 2) * sym.erf(x), x)
  ____     2
\/ pi *erf (x)
--------------
      4
```

It is possible to compute definite integral:

```
>>> sym.integrate(x**3, (x, -1, 1))
0
>>> sym.integrate(sym.sin(x), (x, 0, sym.pi / 2))
1
>>> sym.integrate(sym.cos(x), (x, -sym.pi / 2, sym.pi / 2))
2
```

Also improper integrals are supported as well:

```
>>> sym.integrate(sym.exp(-x), (x, 0, sym.oo))
1
>>> sym.integrate(sym.exp(-x ** 2), (x, -sym.oo, sym.oo))
  ____
\/ pi
```